

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

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$$G(s)H(s) = \frac{k(1+s)}{(s+1)(s+2)}$$

$$= \frac{k(s+1)}{(s+1)(s+1)(s+2)}$$

$$\text{闭环极点: } p_1 = -1, p_2 = -2$$

$$n = 3 \text{ 有 3 根极点.}$$

$$\text{闭环零点: } z_1 = -1, m = 1$$

$$\text{系统类型: } n - m = 2.$$

$$\sigma = \frac{-1-2}{3} = -1.$$

$$\zeta = \frac{2\sigma}{n-m} = 0.7.$$

$$\frac{1}{s+1} + \frac{1}{s+1} + \frac{1}{s+2} = \frac{1}{s+2}$$

$$d = -0.805.$$

$$\alpha = \frac{2\zeta\sigma}{2} = \frac{\sigma}{2}$$

$$\text{特征方程: } s^3 + 3s^2 + 2s + k = 0$$

$$-s^3 - 3s^2 + (2+k)s + 2k = 0$$

$$\begin{cases} -3s^2 + 6 + 2k = 0 & \text{在 } s=0 \\ s(-s^2 - 3s + 2 + k) = 0 & \text{在 } s=-1 \end{cases}$$

无实数解.

根轨迹与虚轴相交于 $\sigma = 0$

取 $p = -1$

$$\text{零点: } -1 - (-1) - (-2) = 1 \quad \theta_2 = 45^\circ$$

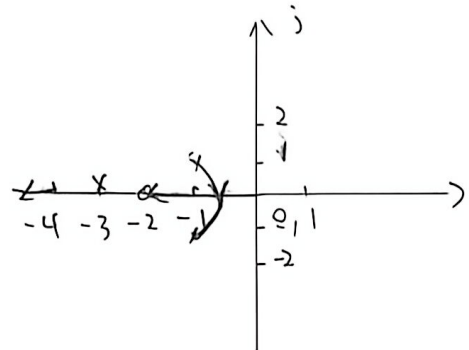
$$\text{极点: } -1 - (-1) - (-3) = 2 \quad \theta_1 = 26.565^\circ$$

$$\text{根轨迹: } -1 - (-1) - (-1) = 2 \quad \theta_2 = 90^\circ$$

$p = -1$ 的角平分线.

$$\theta_1 = 45^\circ - (26.565^\circ + 90^\circ) = -71.565^\circ$$

由于 $p = -1$ 的角平分线 $\theta_2 = 71.565^\circ$



$$\eta_1 = \frac{\omega_n^2}{\zeta^2 + 2\zeta\sigma + \sigma^2} \quad \omega_n = 0.7$$

$$\zeta = 0.309$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 0.7 \sqrt{1 - 0.309^2} = 0.6329$$

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} = \frac{1}{2 \times 0.309 \sqrt{1 - 0.309^2}} = 1.767$$